

Big Data, Big Bias?

A Systematic Review of User-Centric Computational Methods for Collecting Digital Trace Data and Error Correction Methods

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Digital trace data, retrieved via user-centric computational methods such as data donations or tracking, holds great potential for studying social science phenomena. However, the field has yet to introduce standards for defining and using these methods. In particular, social scientists using digital trace data need to be aware of and address errors in representation and measurements related to such data. In this interdisciplinary, preregistered literature review ($N = \text{XXX}$), we use quantitative and qualitative content analysis to (a) map the use of different user-centric CSS methods in the social sciences, (b) analyze the degree to which bias in representation and measurement is made transparent, and (c) outline potential error correction strategies, such as survey design approaches or statistical modeling. Results indicate that XXX . Overall, our study underlines the need to (a) make bias in digital trace data more transparent, (b) rely on such data for answering substantial social science questions instead of purely methodological endeavors, and (c) advance error correction methods.

Keywords: Digital Trace Data, User-Centric Data Collection, Bias, Error, Sample, Measurement, Systematic Review

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